

Notes: Sorensen (JPE, 2000)
Equilibrium Price Dispersion in Retail Markets for Prescription Drugs

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1 Big picture

- **Question.** Why do homogeneous prescription drugs sell for very different cash prices across pharmacies in the same local market?
- **Core hypothesis.** Price dispersion arises in part from costly consumer search. Consumers do not always know every pharmacy's price, and the incentive to search depends on how valuable the price information is.
- **Main empirical setting.** The paper uses posted cash prices for prescription drugs at pharmacies in Middletown and Newburgh, New York. These markets are geographically well defined and contain a moderate number of pharmacies.
- **Key source of identifying variation.** Consumers have stronger incentives to search for drugs that they expect to purchase repeatedly. Search information is more valuable when the same low-price pharmacy can be used multiple times.
- **Main fact.** Cash prices are highly dispersed. The highest posted price for a given prescription is often more than 50 percent above the lowest price, and the average price range is about \$13.17.
- **Main result on dispersion.** Price dispersion is lower for prescriptions with higher purchase frequency. A drug purchased monthly has a substantially smaller price range than a one-time therapy.
- **Main result on markups.** Average absolute margins are also lower for repeatedly purchased drugs, consistent with search intensity reducing both dispersion and markups.
- **Product differentiation check.** Pharmacy fixed effects explain only part of price dispersion. Prices are not simply high at some pharmacies and low at others across all drugs.
- **Cost heterogeneity check.** Differences in wholesale acquisition costs may explain some dispersion, especially for generics, but do not explain the central relationship between purchase frequency and dispersion.
- **Policy implication.** If dispersion reflects costly search, centralizing price information can reduce markups and dispersion, especially for cash-paying consumers.

2 Incremental contribution of the paper

- **Moves equilibrium price dispersion theory to a clean retail setting.** Earlier search models show that price dispersion can be an equilibrium outcome even for homogeneous goods. Sorensen provides direct evidence using posted prices for identical prescription products sold by nearby pharmacies.
- **Uses within-market cross-sectional variation rather than broad cross-market comparisons.** The paper compares drugs within the same local pharmacy markets, which reduces concerns that dispersion is driven by broad market-level differences.
- **Introduces purchase frequency as a search-intensity proxy.** The key empirical idea is that repeatedly purchased prescriptions make price information more durable and valuable, so consumers should search more. This links a measurable drug characteristic to equilibrium price distributions.
- **Jointly tests dispersion and markups.** Search theory predicts that more search should reduce both absolute price dispersion and price-cost margins. The paper tests both predictions and shows that the residual components of dispersion and margins move together.
- **Separates consumer search from pharmacy heterogeneity.** By using pharmacy rankings, pharmacy fixed effects, and residual dispersion measures, the paper shows that a simple product-differentiation story cannot fully account for the observed price patterns.
- **Adds policy relevance to search-cost models.** The paper makes equilibrium price dispersion empirically relevant for price-transparency policy: if consumers face search frictions, information disclosure can discipline prices.

3 Data and measurement

3.1 Data collection and market definition

- New York law required pharmacies to post prices for top-selling prescriptions on state-provided posters.
- Sorensen copied posted cash prices from pharmacies in Middletown and Newburgh, New York, in March 1998.
- The two cities are useful because each has a moderate number of pharmacies and is geographically isolated.
- The analysis drops pharmacies or prescriptions when price posting is stale or incomplete.
- The final data set contains 428 prescription-city observations: 224 in Middletown and 204 in Newburgh.

Interpretation: The data are especially useful because the products are close to homogeneous, prices are posted, and the local choice set is clear. This is much cleaner than comparing prices across distant markets or across differentiated products.

3.2 Price dispersion measures

For prescription i in city j , the main absolute dispersion measures are:

$$\text{Range}_{ij} = \max_p p_{pij} - \min_p p_{pij},$$

where p indexes pharmacies, and the cross-pharmacy standard deviation of price.

The paper also computes residual dispersion measures by first removing drug and pharmacy fixed effects from price and then measuring the range or standard deviation of the residuals.

Interpretation: Absolute dispersion is the main object because many search models treat the cost of obtaining a price quote as a dollar cost rather than a percentage of price. Relative dispersion is checked but is not the primary dependent variable.

3.3 Purchase frequency

Purchase frequency, denoted PFREQ_i , is constructed using dosage and therapy-duration information from *Mosby's GenRx*:

$$\text{PFREQ}_i = \text{expected number of purchases of prescription } i \text{ in one year.}$$

For example, a drug taken once per day for 365 days with 60 tablets per prescription has

$$\text{PFREQ} = (1 \times 365)/60 \approx 6.08.$$

The paper caps PFREQ at 24 so that a small number of very high-frequency prescriptions do not drive the results.

Interpretation: Purchase frequency measures the value of price information. A consumer who finds a low-price pharmacy for a chronic medication can use that information repeatedly before relative prices change.

3.4 Margin measurement

The paper cannot observe exact pharmacy acquisition costs, so it constructs approximate costs using average wholesale prices (AWP):

$$\begin{aligned}\hat{c}_i &= 0.85 \times \text{AWP}_i + 3 \quad \text{for branded drugs,} \\ \hat{c}_i &= 0.70 \times \text{AWP}_i + 3 \quad \text{for generic drugs.}\end{aligned}$$

Average absolute margin is then

$$\text{Margin}_i = \bar{p}_i - \hat{c}_i.$$

Interpretation: The cost measure is imperfect but plausible for testing whether average markups are lower where search incentives are stronger. The key test is not exact cost accounting; it is whether relative patterns in margins line up with the search prediction.

3.5 Controls

The main controls include:

- wholesale cost, to account for the scale of price levels;
- two brand-status dummies: branded with generic competition, and branded without generic competition;
- a Newburgh dummy;
- drug-class dummies for categories of therapy.

Interpretation: These controls address alternative explanations: high-cost drugs may have larger dollar dispersion, branded drugs may have different consumers, the two cities may differ, and users of some drug classes may search more than others.

4 Models and empirical specifications

4.1 Search-theory prediction

The model intuition is:

higher purchase frequency $\Rightarrow$ higher expected benefit from search $\Rightarrow$ more search $\Rightarrow$ lower prices and lower dispersion.

Interpretation: A high-price pharmacy can profit only when some consumers do not know or do not use low-price alternatives. More search reduces the fraction of such consumers.

4.2 Baseline dispersion regression

The main dispersion regression is:

$$\text{Range}_{ij} = \beta_0 + \beta_1 \text{PFREQ}_i + \beta_2 \text{AWP}_i + \beta_3 \text{BR1}_i + \beta_4 \text{BR2}_i + \beta_5 \text{NEWB}_j + \sum_k \gamma_k D_{ik} + \varepsilon_{ij}. \quad (1)$$

- BR1_i indicates branded drugs with generic competition.
- BR2_i indicates branded drugs without generic competition.
- D_{ik} are drug-class indicators.
- The search hypothesis predicts $\beta_1 < 0$.

Interpretation: The coefficient on purchase frequency asks whether drugs with higher search benefits have tighter price distributions, after holding fixed cost, brand status, city, and therapy category.

4.3 Residual dispersion regression

To remove pharmacy heterogeneity, the paper first estimates a price regression with drug and pharmacy effects and then measures the dispersion of the residual prices. The second-stage regression has the same structure:

$$\text{ResidualDispersion}_{ij} = \beta_0 + \beta_1 \text{PFREQ}_i + \beta_2 \text{AWP}_i + \cdots + \varepsilon_{ij}. \quad (2)$$

Interpretation: If purchase frequency still predicts residual dispersion after pharmacy effects are removed, then the result cannot be explained only by some pharmacies being uniformly high quality or high price.

4.4 Margin regression

The average-margin regression is:

$$\text{Margin}_{ij} = \alpha_0 + \alpha_1 \text{PFREQ}_i + \alpha_2 \text{BR1}_i + \alpha_3 \text{BR2}_i + \alpha_4 \text{NEWB}_j + \sum_k \delta_k D_{ik} + u_{ij}. \quad (3)$$

The search hypothesis predicts $\alpha_1 < 0$.

Interpretation: Search theory predicts lower markups, not only lower dispersion. This is an important second test because a purely mechanical dispersion result would not necessarily imply lower average margins.

4.5 Pharmacy fixed-effect decomposition

To assess pharmacy heterogeneity, the paper estimates price regressions with increasing sets of fixed effects:

$$p_{ipj} = \mu_i + \eta_j + e_{ipj}, \quad (4)$$

$$p_{ipj} = \mu_i + \phi_p + \eta_j + e_{ipj}, \quad (5)$$

$$p_{ipj} = \mu_i + \phi_p + \sum_c \phi_{pc} \mathbf{1}\{i \in c\} + \eta_j + e_{ipj}. \quad (6)$$

Interpretation: The increase in R^2 when pharmacy effects are added measures how much of observed dispersion can be attributed to systematic pharmacy-level pricing differences.

5 Identification and empirical design

5.1 What identifies the relationship

- The analysis exploits cross-drug variation in expected repeat purchases.
- The logic is that repeat purchasing increases the private return to learning prices.
- Within a city, the set of pharmacies is fixed, so variation across drugs maps into variation in search incentives rather than variation in market structure.

Interpretation: This is not a randomized experiment, but it is a sharp cross-sectional test of an equilibrium search prediction.

5.2 Why prescription drugs are a useful setting

- Products are highly standardized: a prescription for the same drug and dosage is close to identical across pharmacies.
- Cash prices are posted and legally binding.
- Local markets are geographically compact.
- Many prescriptions differ in expected purchase frequency for reasons tied to medical use rather than pharmacy strategy.

Interpretation: The setting reduces common objections in price-dispersion studies, such as quality differences, hidden price menus, and unclear consumer choice sets.

5.3 Main identification threats

- **Drug-class heterogeneity:** users of different drugs may have different search propensities for reasons unrelated to repeat purchasing.
- **Pharmacy heterogeneity:** some pharmacies may offer better service, convenience, or location and therefore charge higher prices.
- **Wholesale cost dispersion:** especially for generics, different pharmacies may face different acquisition costs.
- **Insurance coverage:** posted cash prices matter mostly for uninsured or cash-paying consumers, not insured consumers with fixed copayments.
- **Static price snapshot:** the data are cross-sectional, so the analysis cannot directly observe consumer search or price changes over time.

Interpretation: The paper addresses these threats with controls, residual dispersion measures, pharmacy fixed effects, brand/generic comparisons, and margin analysis. The evidence is strongest when the same pattern appears across several measures.

6 Tables and figures

6.1 Table 1: Price Rankings by Pharmacy

Read:

- The table groups each pharmacy's prices into low-, middle-, and high-price categories across prescriptions.
- In Middletown, no pharmacy is uniformly the lowest price. Even stores that are often low-price have many middle-price observations.
- Rite-Aid stores are consistently more likely to be in the highest-price group, especially in both Middletown and Newburgh.
- In Newburgh, Price Chopper and Wal-Mart are often in the lowest-price group, but even there the cheapest provider varies by drug.
- The table visually rejects a simple story in which one fixed pharmacy-quality ranking determines all price differences.

Economic message: The consumer cannot cheaply infer the lowest-price pharmacy from a stable pharmacy ranking. This makes search valuable and supports the interpretation that price dispersion reflects information frictions rather than only product differentiation.

TABLE 1
PRICE RANKINGS BY PHARMACY
A. MIDDLETOWN

PHARMACY	PRICE GROUP		
	Lowest 3	Middle 4	Highest 3
Eckerd	45	103	10
Eckerd	29	102	27
Immediate	43	54	61
K-Mart	56	57	45
Medicine Shoppe	99	49	10
Price Chopper	80	67	11
Rite-Aid	3	11	144
Rite-Aid	2	18	138
Rx Place	38	104	16
Wal-Mart	79	67	12

B. NEWBURGH

PHARMACY	PRICE GROUP		
	Lowest 3	Middle 3	Highest 3
Ace	26	112	30
Hudson	33	106	29
Medical Arts	73	65	30
Price Chopper	134	27	7
Rite-Aid	4	23	141
Rite-Aid	10	45	113
Rite-Aid	18	34	116
Rx Place	64	70	34
Wal-Mart	142	22	4

Table 1: Price Rankings by Pharmacy

6.2 Table 2 and Table 3: Price Dispersion and Purchase Frequency; Average Margins and Purchase Frequency

Read: Table 2: Price Dispersion and Purchase Frequency

- The coefficient on purchase frequency is negative in all four dispersion specifications.
- In column 1, the coefficient is -0.336 for the price range. This implies that repeated purchasing substantially narrows the dollar price range.
- In column 2, the standard deviation coefficient is -0.173 , showing that the result is not driven only by extremes of the price distribution.
- In columns 3 and 4, the residual dispersion measures remain negative after removing pharmacy effects.
- Wholesale cost is positively related to dispersion, consistent with dollar dispersion being larger for more expensive drugs.

Read: Table 3: Average Margins and Purchase Frequency

- In column 1, purchase frequency has a coefficient of -0.262 on average absolute margins.
- The paper interprets this as monthly prescriptions having much lower margins than one-time prescriptions.
- The average-price specification in column 2 is also negative, although weaker statistically.

- The relative-margin specification in column 3 is close to zero, which is consistent with the paper’s focus on absolute search costs and absolute margins.
- Residual correlations are high across cities, suggesting common drug-level components in search intensity and pricing.

Economic message: These two tables are the core empirical test. More valuable search information is associated with both less dispersion and lower absolute markups. That joint pattern is exactly what equilibrium search models predict.

TABLE 2
PRICE DISPERSION AND PURCHASE FREQUENCY

	DISPERSION MEASURE			
	Range (1)	Standard Deviation (2)	Residual Range (3)	Residual Standard Deviation (4)
Purchase frequency	-.336 (.123)	-.173 (.076)	-.266 (.061)	-.102 (.016)
Wholesale cost	.280 (.033)	.180 (.020)	.215 (.043)	.069 (.014)
Branded with generic competition	-.803 (1.037)	-1.480 (.641)	-1.842 (.861)	-.362 (.248)
Branded without ge- neric competition	-1.505 (2.108)	-2.010 (1.303)	-1.967 (1.060)	-.772 (.339)
Newburgh dummy	-2.686 (.633)	-3.172 (.314)	-1.493 (.791)	-.916 (.271)
Constant	20.070 (4.343)	7.321 (2.563)	14.570 (1.062)	5.283 (.448)
R^2	.371	.447	.258	.253
$\hat{\rho}$	.338	.585	.149	.648

NOTE.—GLS estimates allowing for correlation in the error terms across cities for each prescription ($\hat{\rho}$ is the estimated correlation); standard errors are in parentheses. The number of observations is 428. The residual range (standard deviation) is the range (standard deviation) of the residuals from a regression of price on drug and pharmacy fixed effects, as described in the text. Estimated coefficients for a set of 20 drug class dummies are suppressed; a table listing the full set of coefficients is available from the author on request.

(a) Table 2: Price Dispersion and Purchase Frequency

TABLE 3
AVERAGE MARGINS AND PURCHASE FREQUENCY

	DEPENDENT VARIABLE		
	Average Margin (1)	Average Price (2)	Average Relative Margin (3)
Purchase frequency	-.262 (.102)	-.137 (.105)	.001 (.003)
Wholesale cost	...	.994 (.032)	...
Wholesale cost $\times$ generic dummy	...	-.208 (.059)	...
Branded with generic competition	2.101 (.720)	-.668 (1.056)	-.235 (.020)
Branded without generic competition	3.415 (1.660)	-.123 (1.891)	-.255 (.046)
Newburgh dummy	1.681 (.174)	1.648 (.140)	.047 (.005)
Constant	12.69 (2.435)	11.86 (2.581)	.463 (.068)
R^2	.229	.895	.510
$\hat{\rho}$	.915	.936	.898

NOTE.—GLS estimates allowing for correlation in the error terms across cities for each prescription ($\hat{\rho}$ is the estimated correlation); standard errors are in parentheses. The number of observations is 428. Estimated coefficients for a set of

(b) Table 3: Average Margins and Purchase Frequency

6.3 Table 4: Explaining Price Variation with Pharmacy Fixed Effects

Read:

- Drug fixed effects alone explain most price levels, with $R^2 = 0.907$, because drugs differ greatly in average price.
- Adding pharmacy fixed effects raises R^2 to 0.938 and reduces the mean absolute residual from 3.47 to 2.74.
- This means pharmacy effects explain part, but not most, of the remaining within-drug price dispersion.
- Interacting pharmacy effects with drug type only raises R^2 to 0.942 and further reduces mean residuals to 2.51.
- The limited incremental fit from interactions suggests that pharmacies are not simply high quality for some drug categories and low quality for others.

Economic message: Pharmacy heterogeneity matters, but it is insufficient. Even after allowing pharmacies to have systematic price differences, meaningful residual dispersion remains, and the search-based purchase-frequency pattern persists.

TABLE 4
EXPLAINING PRICE VARIATION WITH PHARMACY FIXED EFFECTS

REGRESSOR	$\overline{R}^2$ (1)	$ \epsilon $	
		Mean (2)	Standard Deviation (3)
Drug effects	.907	3.47	3.78
Drug and pharmacy effects	.938	2.74	3.13
Drug and pharmacy effects, with drug type interactions	.942	2.51	2.86

NOTE.—Based on regressions with price as the dependent variable. Each regression also includes a city dummy and a generic dummy. The means and standard deviations pertain to the absolute magnitudes of the regression residuals. The F -statistic testing the hypothesis that the pharmacy effects are all jointly zero in the second regression is 112.31 ($p = .000$); the F -statistic for the interaction terms in the third regression is 1.535 ($p = .000$).

Table 4: Explaining Price Variation with Pharmacy Fixed Effects

7 Short summary

- The paper studies cash prices for prescription drugs in two local New York markets.
- Posted prices for the same prescription differ substantially across nearby pharmacies.
- The main explanation tested is costly consumer search.
- The key proxy for search incentives is purchase frequency: repeated purchases make price information more valuable.
- Price ranges and price standard deviations are lower for drugs purchased repeatedly.
- The same negative relationship holds after removing pharmacy fixed effects.
- Average absolute margins are also lower for frequently purchased drugs.
- Pharmacy fixed effects explain roughly one-third of the observed residual price dispersion but do not eliminate it.
- Cost heterogeneity, especially for generics, may explain some dispersion but not the main purchase-frequency pattern.
- The paper’s evidence supports equilibrium search models in which price dispersion is a stable outcome when consumers have imperfect price information.

8 Conclusion

8.1 Core conclusion

Sorensen shows that retail price dispersion for prescription drugs is large and systematic. The dispersion is not merely noise, nor is it fully explained by persistent pharmacy-level price differences. Instead, the cross-drug pattern is consistent with equilibrium consumer search: drugs for which consumers have stronger incentives to search have lower dispersion and lower margins.

8.2 Incremental contribution revisited

The paper’s incremental contribution is to take the equilibrium price dispersion literature out of purely theoretical settings and test it in a retail market where product homogeneity and local market boundaries are unusually clear. The empirical design is simple but powerful: use purchase frequency to proxy for the

durability and value of price information. This gives the paper a direct link between a core state variable in search models and observable price distributions.

8.3 Economic intuition

Search costs allow different pharmacies to charge different prices for the same prescription. A high-price pharmacy can survive because some consumers either do not know the low price or decide the expected savings do not justify search. For chronic or frequently purchased drugs, the same search effort yields repeated savings. Consumers therefore search more, and pharmacies respond by keeping prices closer together and margins lower.

8.4 What the paper teaches about policy

If price dispersion comes from search frictions, price transparency can improve consumer welfare. However, the effect will be strongest for cash-paying consumers and for drugs where consumers can substitute across pharmacies. The paper therefore speaks directly to policies that centralize pharmacy price information, especially for uninsured consumers or those with high deductibles.

8.5 Limitations and open questions

The paper is based on a cross-section of posted prices in two cities, so it cannot directly observe individual search behavior, the dynamics of price changes, or how consumers update beliefs after searching. The cost measure is also approximate, relying on average wholesale prices and assumed discounts. Finally, the setting is prescription drugs in two local markets; the same mechanism should travel to other markets, but the size of the effect may depend on product complexity, consumer demographics, online price transparency, and the persistence of relative prices.

8.6 Takeaway

The durable takeaway is that price dispersion can be an equilibrium phenomenon caused by costly information acquisition, not simply a measurement error or a sign of product differentiation. Sorensen's evidence shows that when consumers have more reason to search, dispersion and markups fall. This is the empirical signature that ties retail price patterns to search theory.